YIFEI SUN

Staff Software Engineer · Distributed Systems Orchestration & Cloud-Native Infrastructure · 13 Years

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SUMMARY

Staff-level infrastructure engineer with 13 years building distributed systems orchestration, cloud-native platform services, and high-scale API servers in Java. Led cross-team redesign of a high-traffic orchestration layer serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%), consolidating 12 downstream services into 5 via a new orchestration service with lifecycle management, fault isolation, and controlled rollout at 50% production traffic. Established architecture standards across 10+ teams and 3 departments. Built service orchestration for AI agent workflows (MCP server) and event-driven microservice platforms on Kubernetes. Early engineer at Toast through hypergrowth; combines deep hands-on Java engineering with cross-organizational technical leadership.

LEADERSHIP HIGHLIGHTS

- Led cross-team orchestration layer redesign with lifecycle management and controlled rollout: p99 5× reduction, 99.9% availability
- Drove architecture and observability standards across 10+ teams, 3 departments via RFCs and design docs
- Built service orchestration for AI agent workflows (MCP server) and event-driven microservice platforms on Kubernetes
- Mentored senior engineers to Principal-level scope; ships 0→1 cloud-native services fast with measurable impact

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), Python · AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL · Service Orchestration, Control Plane Design, API Servers · Microservices, Event-Driven, CQRS · Datadog, Grafana, Distributed Tracing · CI/CD, Container Orchestration

INDEPENDENT PROJECT

Distributed Orchestration & Evaluation Platform

- Architected a multi-path orchestration engine with configurable execution strategies, lifecycle management (submission, execution, recovery), and full observability across heterogeneous downstream services.
- Designed a scheduler-driven control loop with DB-backed configuration, environment-specific overrides, and automated parameter optimization — enabling safe, iterative rollout of orchestration policy changes.
- Built an evaluation system for automated quality assessment, offline backtesting, and parameter search (grid & Bayesian optimization) across multiple execution providers.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Traffic API Orchestration Layer — Cross-Org Platform Migration

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core orchestration layer behind the most-visited page (~8,000 rpm, millions of daily impressions); managed full lifecycle from design through controlled rollout at 50% production traffic.
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined services with lifecycle management, fault isolation, circuit breaking, and graceful degradation.
- Built a parity-testing framework that automatically compared new vs. legacy responses across full payload and data-integrity dimensions, enabling safe, data-driven rollout decisions.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards and observability requirements for new services; instrumented with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into a federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.

- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.

MCP Server — Service Orchestration for AI Agent Workflows (0→1)

- Sole backend engineer designing and implementing an MCP server for AI agent orchestration, exposing platform capabilities as callable services with lifecycle management, request routing, and fault handling.
- Designed the orchestration layer for multi-step agent workflows with configurable execution policies, timeout management, and safe incremental rollout alongside existing API infrastructure.

Digital Shopping & Commerce Platform (0→1)

- Led backend design for a new digital commerce product launched during the pandemic, using microservice and event-driven architecture for inventory, merchandising, and lead flows on Kubernetes.
- Built infrastructure for rapid product experimentation — enabling the company to quickly launch, measure, and safely retire product bets without destabilizing core platform services.

Cross-Org Platform Leadership & Architecture Standards

- Established patterns for API design, service orchestration, caching, and distributed fault handling adopted across multiple teams; authored design docs, RFCs, and decision records driving consensus.
- Mentored senior engineers on distributed systems design, service lifecycle management, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010